09 - Modeling Customer Choice

IEOR 145/245

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So far…

- We **assign/allocate** resources
  - Demand can be uncertain over time
  - Suitable for Ads and school bus problem

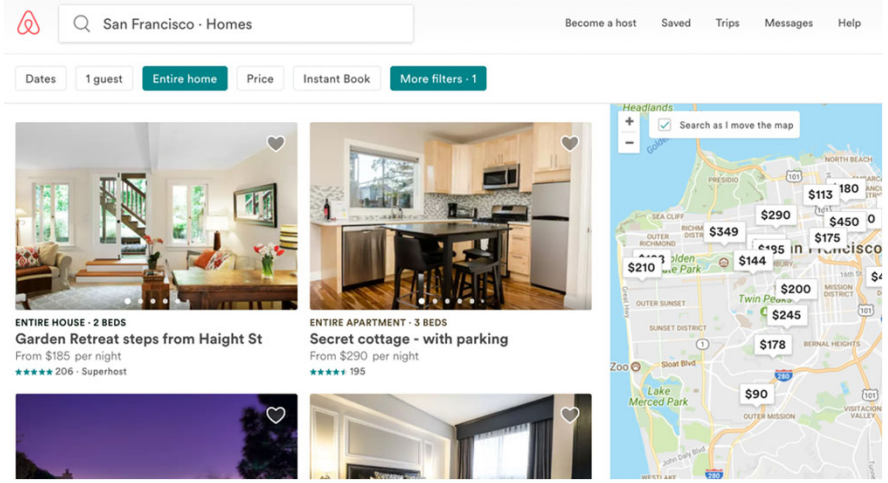
Module 2: Uncertain Customer Choice

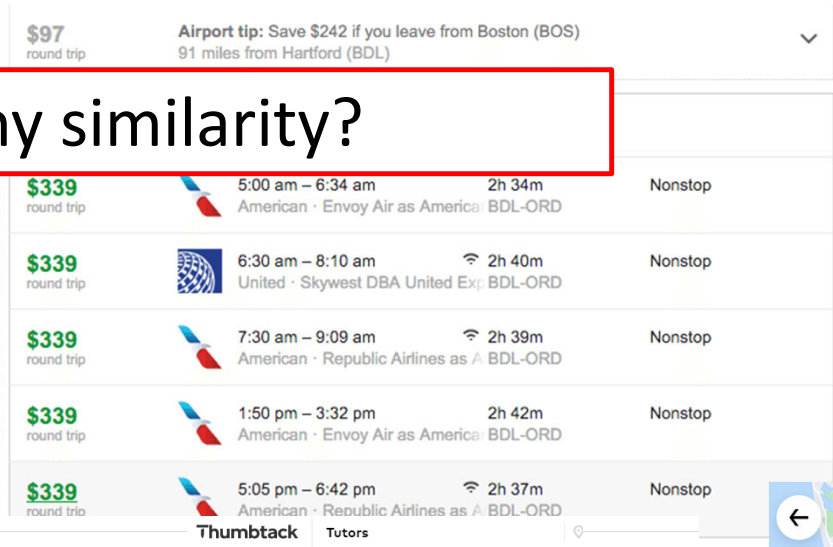
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  - Demand can be uncertain over time
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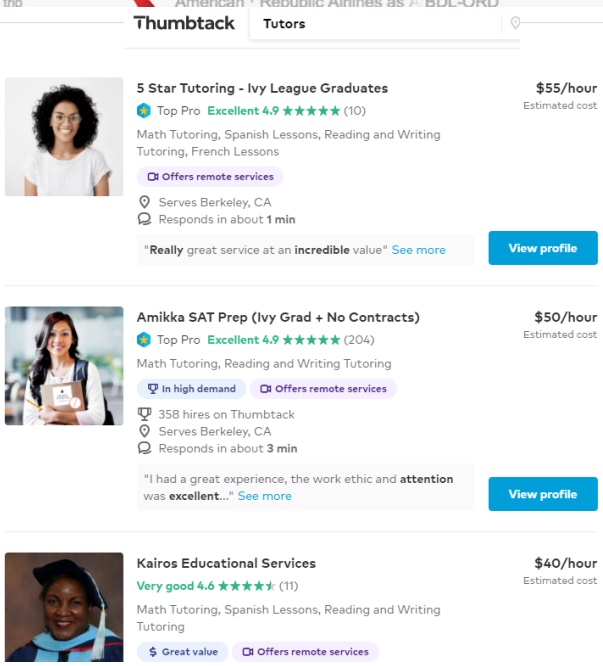
- In many settings, such as airlines and e-commerce, customers *choose* and we only decide a *set of options* to display
  - How do we model customer choice?
  - How do we show the right options?

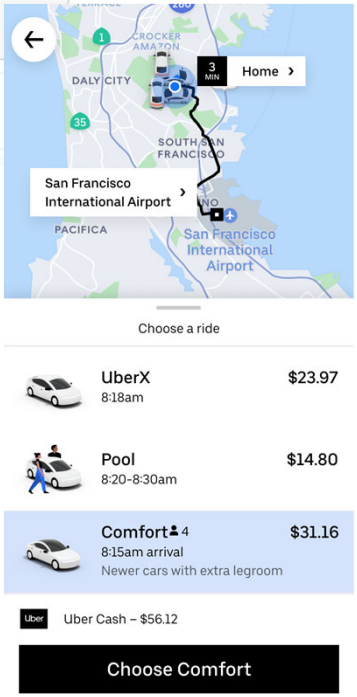
Notice any similarity?





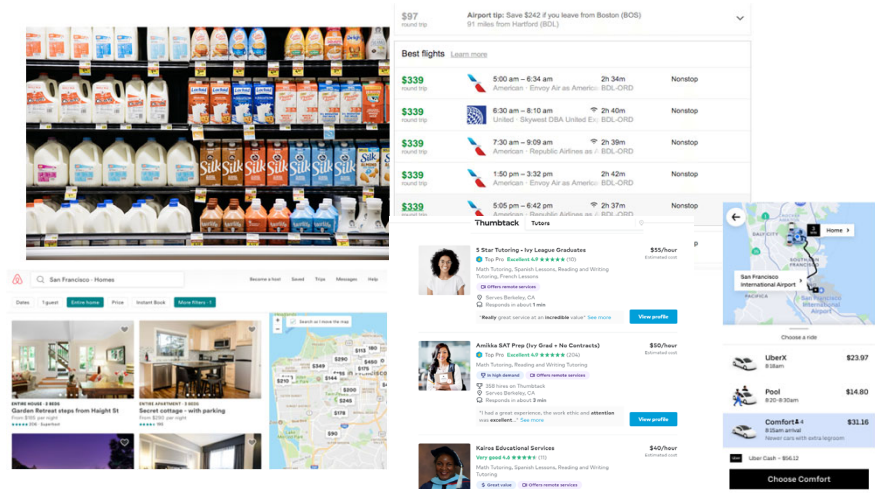






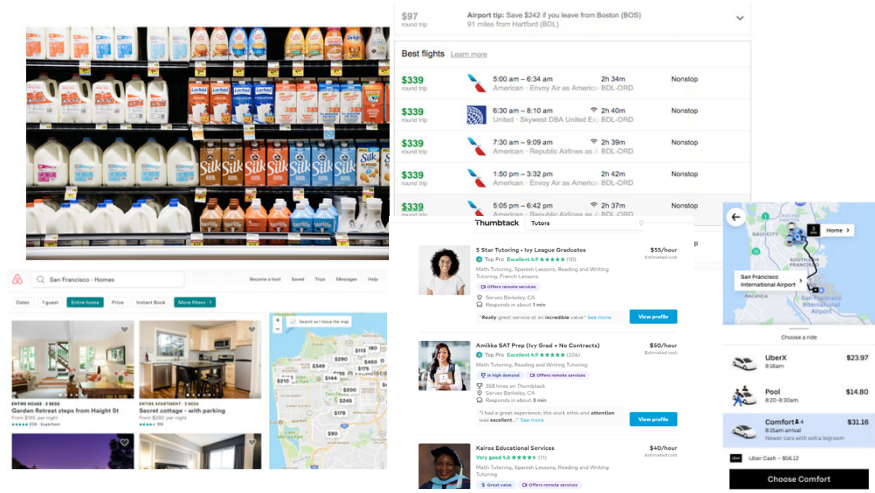
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  - We are picking *between* them
  - Involves comparison



What is common in these applications?

- Products are **substitutable**
  - We are picking *between* them
  - Involves comparison
- Some situations involve **complementary** products
  - E.g., milk and cereal, flight and hotel
  - We treat these choices as independent
- We focus on substitutable goods/products/items



How do you choose?



How do you choose?

- Pick cheapest fat-free
- Pick cheapest organic
- Pick most expensive
- Pick specific brand and type ...

Very hard to predict



How does a population choose?

How does a population choose?

- Probability distribution over items
- This distribution is called a **Choice Model**
- We will see two equivalent ways of modeling choice
  1. Ranking choice model
  2. Random utility model

Choice Models

Thought Experiment



$p_1 = 0.3$



$p_2 = 0.4$



$p_3 = 0.2$

None of these

$p_0 = 0.1$

- (Probability × 100) is the % of population who would buy the shoe

Thought Experiment



$p_1 = 0.3$?



$p_2 = 0.4$?



$p_3 = 0.2$?

None of these

$p_0 = 0.1$?

- We add a new option,

  what do we expect the probabilities to look like?

New option



$p_4 =$?

Thought Experiment



$p_1 = 0.3$



$p_2 = 0.3$



$p_3 = 0.1$

None of these

$p_0 = 0.09$

- Fewer people will buy the older options
- **Probabilities should decrease**
  - Probabilities depend on the **set** of items

New option



$p_4 = 0.21$

Choice Model

- Describes how people choose at most one product from an assortment

Choice model ϕ: Given set of options (assortment) S,

$\phi(i, S) =$ probability that a random customer chooses product i from S

alternatively, fraction of population that chooses i from S

Basic Properties of Probabilities

- $\phi(i, S) \in [0,1]$
- $\phi(S) \coloneqq \sum_{i \in S} \phi(i, S) =$ Probability that customer chooses something
- $\phi(S) + \phi(0, S) = 1$
  - $\phi(0, S)$ is the probability of choosing nothing
  - 0 is also called **no purchase or outside option**

Choice Model: Examples

- **Shoes:** $S=\{1,2,3\}$
  - $\phi(1,S)=0.3$: 30% of shoppers pick shoe 1;
  - Suppose $\phi(2,S)=0.4$ and $\phi(3,S)=0.2$, then
    - $\phi(S)$ adds up the choice probabilities of all products in S
    - $\phi(S)=0.3+0.4+0.2=0.9$ → 90% buy some shoe
  - Everyone else picks the outside option
    - $\phi(0,S)=1-0.9=0.1$ → 10% buy nothing

- **Ride-hailing**: $S=$ {UberX, Pool, Comfort}
  - $\phi(\text{Comfort},S)$ = fraction of riders who request Comfort
  - Outside option 0: take the bus or walk

Important Properties of Choice Models

1. **Substitutability:** Adding a new option never increases the probability of selection of other options

$$\phi(i, S \cup \{j\}) \leq \phi(i, S) \qquad \forall j \notin S, i \in S$$

(new option)

- Items/products are substitutes

- For complementary items, new option may enhance probabilities
  - E.g., milk and cereal
  - Not captured by choice models

Important Properties of Choice Models

1. Substitutability: Adding a new option never increases the probability of selection of other options

$$\phi(i, S \cup \{j\}) \leq \phi(i, S) \quad \forall j \notin S, i \in S$$

2. **Monotonicity:** Adding a new option increases overall probability of selection

$$\phi(S \cup \{j\}) \geq \phi(S) \quad \forall j \notin S$$

means for every product j that is not already offered in S

Properties in Action: Adding Shoe 4

- Before: $S=\{1,2,3\}$. After adding shoe 4: $S\cup\{4\}=\{1,2,3,4\}$
- **Substitutability**: no old product gains
  - Shoe 1: 0.3 → 0.3 Shoe 2: 0.4 → 0.3 Shoe 3: 0.2 → 0.1

- **Monotonicity**: total purchases do not drop
  - $\phi(S)=0.9 \rightarrow \phi(S\cup\{4\})=0.3+0.3+0.1+0.21=0.91$
  - Shoe 4 steals 0.2 from shoes 2 and 3, and wins 0.01 new buyers who used to walk away

1. Ranking Choice Model (RCM)

Main Idea

- Every customer shortlists then picks best available on some criteria

- **Customer 1:** shortlists organic cow milk
  - Picks cheapest option provided ≤ $7

- **Customer 2:** shortlists “Silk” brand
  - Picks lowest fat milk

- **Customer 3:** looking for Oatly brand
  - Unavailable, leaves store

- **Customer 4:** same preference as customer 1



Main Idea

- Every customer shortlists then picks best available on some criteria

Shortlisted products, 0 (outside options), other products

→ Worse rank

E.g.: {clover 2%, oatly, 0, other options}

Every customer has a ranking of products and outside option



Ranking Choice Model (RCM)

- Customers **rank order products and outside option** and pick best available option
  - Outside option always available

- Given a "universe" of n products, $(n+1)!$ rank orders
  - All possible ways of permuting n products +1 outside option
  - Example with 2 products, 1 2 0, 0 1 2, 1 0 2, 2 1 0, 0 2 1, 2 0 1

- Choice model is a **distribution over ranked order** list of products

Interpreting Rank Orders

- A rank order is a customer's list with best first
  - 1 2 0: “I want 1; if it’s not there, 2; otherwise I leave”
  - 2 0 1: “I only want 2; I'd rather leave than buy 1”
  - 0 1 2: “I'm just browsing, I buy nothing”

- E.g., coffee shop with 1 = latte, 2 = cold brew
  - A 1 2 0 customer who finds lattes sold out buys cold brew

- The number of lists $(n+1)!$ grows fast
  - $n=2$: 3! = 6 $n=3$: 4! = 24 $n=10$: 11! ≈ 40 million

RCM – Distribution over Rank Orders

- Customer has rank order σ with probability λ_σ
  - Θ is set of all rank orders
  - For example, $\Theta = \{1\ 2\ 0, 0\ 1\ 2, 1\ 0\ 2, 2\ 1\ 0, 0\ 2\ 1, 2\ 0\ 1\}$

    $\lambda_{1\,2\,0} = 0.1$ $\lambda_{2\,0\,1} = 0.9$
  - λ_σ: probability that customer has rank order σ
  - $\sum_{\sigma \in \Theta} \lambda_\sigma = 1$

RCM – Choice Probabilities

- What is $\phi(i, S)$?

- If we offer assortment S,
  - Customer has rank order σ w.p. λ_σ
  - Chooses top ranked product in $S \cup \{0\}$ as given by σ

RCM – Choice Probabilities

- What is $\phi(i, S)$?

- If we offer assortment S,
  - Customer has rank order σ w.p. λ_σ
  - Chooses top ranked product in $S \cup \{0\}$ as given by σ

$$\phi(i, S) = \sum_{\sigma \in \Theta} \lambda_\sigma \; \mathbb{I}_\sigma(i \text{ top ranked in } S \cup \{0\})$$

0/1 indicator that i is top ranked for customers with rank order σ

RCM: Computing Choice Probabilities

- Recipe for $\phi(i,S)$
  - **Step 1:** in each rank order σ, cross out products not in S
  - **Step 2:** the first remaining entry is what that customer picks
  - **Step 3:** add up λ_σ over the rank orders that pick i

- Example: $\lambda_{1\,2\,0}=0.1$, $\lambda_{2\,0\,1}=0.9$, offer $S=\{1\}$
  - 1 2 0 → 1 0 → picks 1; 2 0 1 → 0 1 → leaves
  - $\phi(1,\{1\})=0.1\times1+0.9\times0=0.1$

Example

- 2 products, $\Theta = \{1\ 2\ 0, 0\ 1\ 2, 1\ 0\ 2, 2\ 1\ 0, 0\ 2\ 1, 2\ 0\ 1\}$
- Define RCM through λ_σ: $\{0.5,\ \ 0,\ \ 0,\ \ 0,\ \ 0,\ \ 0.5\}$
  - i.e., $\lambda_{1\,2\,0} = \lambda_{2\,0\,1} = 0.5$ and all other rank orders have probability 0
- Given set $S = \{2\}$ i.e., only item 2 is available,

  customer chooses 2 with probability $\phi(2, \{2\}) = 1$
- Given $S = \{1\}$,

  $\phi(1, \{1\}) = 0.5$
- Given $S = \{1,2\}$?

Example

- 2 products, $\Theta = \{1\ 2\ 0, 0\ 1\ 2, 1\ 0\ 2, 2\ 1\ 0, 0\ 2\ 1, 2\ 0\ 1\}$
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- Given $S = \{1,2\}$,

  $\phi(1, \{1,2\}) = \phi(2, \{1,2\}) = 0.5$ and $\phi(\{1,2\}) = 1$

2. Random Utility Model (RUM)

Random Utility Model (RUM)

Idea: Customers have utility for each product and
pick product with highest positive utility

- Utility captures value – cost, and possibly other factors
- Outside option has utility 0
- Some products may have negative utility (cost more than value)

Utility: An Everyday Example

- Lunch near campus: utility = value to you − price
  - Burrito: worth $12, costs $10 → utility +2
  - Poke bowl: worth $15, costs $14 → utility +1
  - Salad: worth $9, costs $11 → utility −2
  - Skip lunch → utility 0

- You pick the **burrito**: highest positive utility

- If only the salad is offered, you skip lunch ($-2 < 0$)

Random Utility Model (RUM)

- Utility is a random variable $u_i \; \forall i \in [n]$
  - Outside option always has 0 utility

- RUM is specified by a joint distribution over utilities $(u_i)_{i \in I}$
  - Joint distribution allows correlations between utilities

- Given assortment S, customer chooses item with highest positive utility
  - Random utilities → random choice
  - If all items in S have negative utility → outside option

Why Random? Why Joint?

- **Random**: different customers value products differently
  - Distribution over population captured via random utilities

- **Joint distribution**: utilities of related products can move together
  - A customer who loves Coke probably okay with Pepsi

Typical Decomposition of Utility

Random variable: $u_i = c_i - \alpha\, r_i + \epsilon_i$

- c_i denotes the mean value of item i
- r_i denotes the price of the item
  - $c_i - \alpha\, r_i$ is the mean utility of the item
- ϵ_i is random (with mean 0)

Choice Probabilities in RUM

- Given an assortment $S \subseteq [n]$, customer with utilities $(u_i)_{i \in [n]}$ chooses

$$i^* = \operatorname{argmax}_{i \in S \cup \{0\}} u_i$$

where $u_0 = 0$.

$$\phi(i, S) = ?$$

argmax = option with the largest value (the option, not the value)

Choice Probabilities in RUM

- Given an assortment $S \subseteq [n]$, customer with utilities $(u_i)_{i\in[n]}$ chooses

$$i^* = \operatorname{argmax}_{i\in S\cup\{0\}} u_i$$

where $u_0 = 0$.

$$\phi(i,S) = \Pr(\, i = i^*) = \Pr(u_i = \max_{j\in S\cup\{0\}} u_j)$$

argmax = option with the largest value (the option, not the value)

Example of RUM

- 2 products
  - $u_1 = 1$ with probability (w.p.) 0.5
  - $u_1 = -1$ w.p. 0.5
  - $u_2 = 0.5$ w.p. 1

- Given set $S = \{2\}$ i.e., only item 2 is available,

$$\phi(2, \{2\}) = 1$$

- Given $S = \{1\}$,

$$\phi(1, \{1\}) = 0.5$$

- Given $S = \{1,2\}$?

Example of RUM

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- Given set $S = \{2\}$ i.e., only item 2 is available,

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- Given $S = \{1\}$,

$$\phi(1, \{1\}) = 0.5$$

- Given $S = \{1,2\}$,

$$\phi(1, \{1,2\}) = \phi(2, \{1,2\}) = 0.5 \text{ and } \phi(\{1,2\}) = 1$$

Fact: RCM = RUM

- Intuitively, both models are equally expressive/powerful
  - One thinks in terms of lists, another in terms of sorting utilities

- Formally, given a RCM (ranking model), we can define a RUM (utility model) such that,

$$\phi_{RCM}(i, S) = \phi_{RUM}(i, S) \quad \forall\, i \in S, S \subseteq [n],$$

and vice-versa.

A Forest of Choice Models

